

- 3 (a) Young modulus = stress / strain B1 [1]
- (b) (i) $E = (F \times l) / (A \times e)$ or $e = (F \times l) / (A \times E)$ B1
- $e \propto 1/E$
or
ratio $e_c / e_s = E_s / E_c$ or $(1.9 \times 10^{11}) / (1.2 \times 10^{11})$ or 19/12 C1
- (ratio =) 1.6 (1.58) A1 [3]
- (ii) two straight lines from (0,0) with **S** having the steepest gradient B1 [1]